IMDB Neural Network Model Summary Report

This report summarizes results of nine neural network experiments used to classify IMDB movie reviews as positive or negative.

Executive summary

Main finding is that network with two hidden layers and 64 units in each layer generates the strongest validation result. It reached 88.78% validation accuracy and 88.16% test accuracy. Original class model reached 88.60% validation accuracy and 88.06% test accuracy. Wider model performed best, but its advantage was modest rather than dramatic.

Experiments also showed that, more complicated model is not always better. One hidden layer performed almost as well as the selected model, while three layers produced only minor improvement over baseline. Binary crossentropy worked better than mean squared error for this classification task. Tested dropout and L2 settings did not improve validation accuracy.

9	2 x 64 ReLU	88.78%	88.16%
Models tested	**Selected model**	**Validation accuracy**	**Test accuracy**

Recommendation

Use 2 x 64 ReLU model as final model because it ranked first on validation accuracy and also improved test accuracy slightly. Keep conclusion measured: experiment supports this choice under the current data split and training settings, but it does not prove that 64 units will always be best.

Question and modeling approach

What the model is trying to do:

Task is binary classification. Each movie review must be placed into one of two categories: negative or positive. Model produces a number between 0 and 1. Values closer to 1 indicate a stronger prediction that the review is positive, while values closer to 0 indicate negative prediction.

Data used:

IMDB dataset contains 50,000 labeled movie reviews. Original data is evenly divided between 25,000 training reviews and 25,000 test reviews. From original training set, 10,000 reviews were reserved for validation and 15,000 were used to update model weights. Test set remained separate so it could provide final check on unseen data.

Each review was converted into a 10,000 position numeric vector based on most frequent words in dataset. Value of 1 indicates that a word appeared in the review, and 0 indicates that it did not. This representation is called multi hot encoding. It gives neural network a fixed size numeric input, although it does not preserve word order.

Procedure:

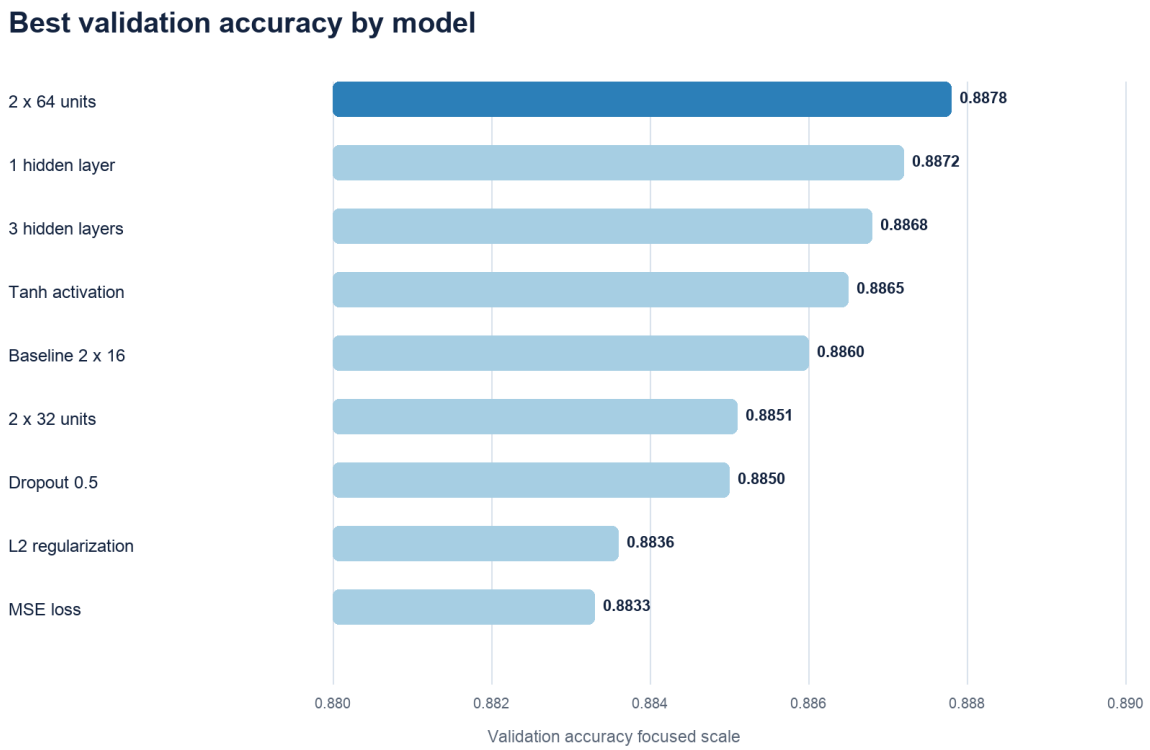
Step	Action	Purpose
1	**Prepare the data**	Load the IMDB reviews, convert them into numeric vectors, and create training, validation, and test sets.
2	**Build the baseline**	Use two 16-unit hidden layers, ReLU activation, binary crossentropy, RMSprop, and a sigmoid output.
3	**Change one choice**	Compare depth, hidden units, loss, activation, dropout, and L2 regularization.
4	**Train consistently**	Use the same split, seed, batch size, optimizer, and maximum epoch count for every experiment.
5	**Select and verify**	Rank models by validation accuracy, then report performance on the separate test set.

How performance was judged:

Accuracy is the percentage of reviews classified correctly. Training accuracy is performance on reviews used for learning. Validation accuracy defines as performance on reviews held back while model choices are being compared. Test accuracy provides final estimate on a separate set of reviews. Validation

performance was the main basis for choosing final model because selecting models directly from test results would make test set less independent.

Overall findings



Best validation accuracy for all nine models

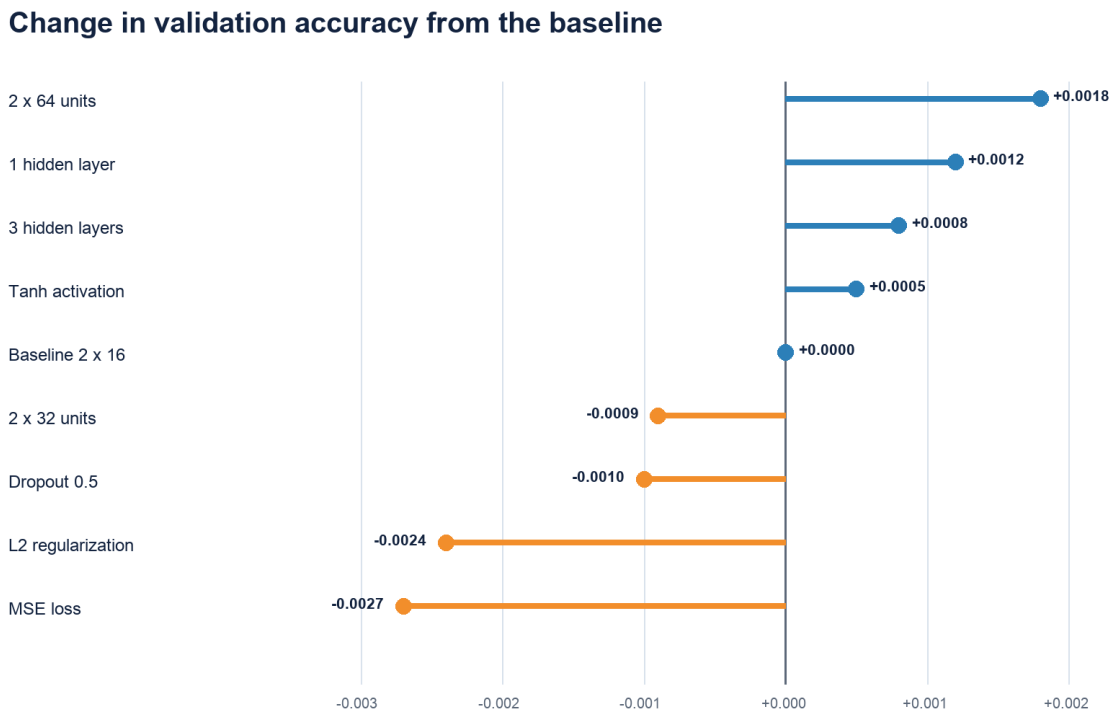
Ranking shows that 2 x 64 model had highest validation accuracy at 0.8878. One layer model followed at 0.8872, and three layer model reached 0.8868. Baseline ranked fifth at 0.8860. Focused horizontal scale is intentional because all models performed within less than half of one percentage point of one another.

Differences are meaningful for comparison but not large enough to claim a breakthrough. Selected model correctly classified about 88 out of every 100 validation reviews. Its lead over baseline was 0.18 percentage points, equal to roughly 18 additional correct classifications per 10,000 reviews if the same difference held in similar data.

What this means:

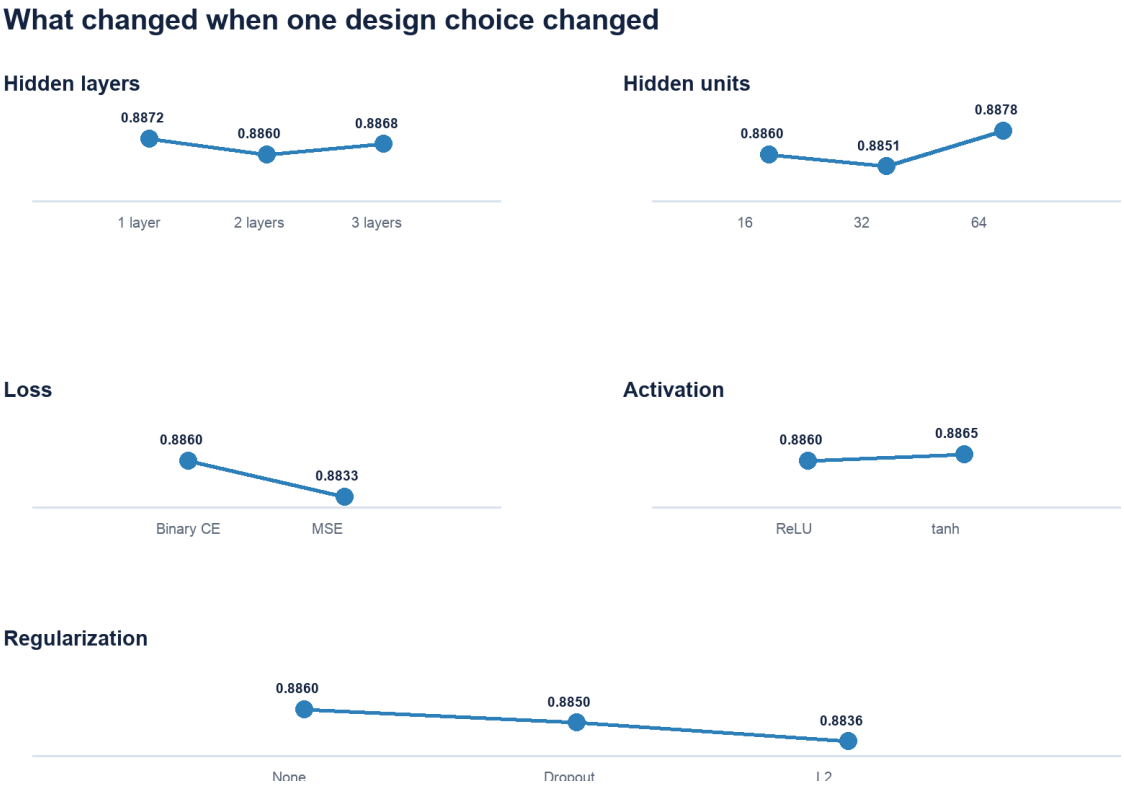
Wider network helped more than deeper network in this experiment. However, one layer model remained highly competitive. If speed, memory use, or simplicity were the highest priority, one layer model could be considered because it achieved nearly same validation score with simpler architecture. For this assignment, where validation accuracy is part of grading criteria, the 2 x 64 model remains strongest choice.

Effect of each modeling decision:



Change in validation accuracy compared to baseline

Four configurations exceeded baseline: 64-unit layers, one hidden layer, three hidden layers, and tanh activation. Largest gain came from 64-unit model. 32 unit model, dropout model, L2 model, and MSE model were below the baseline. This result is useful because it shows that improvements were not proportional to model complexity.

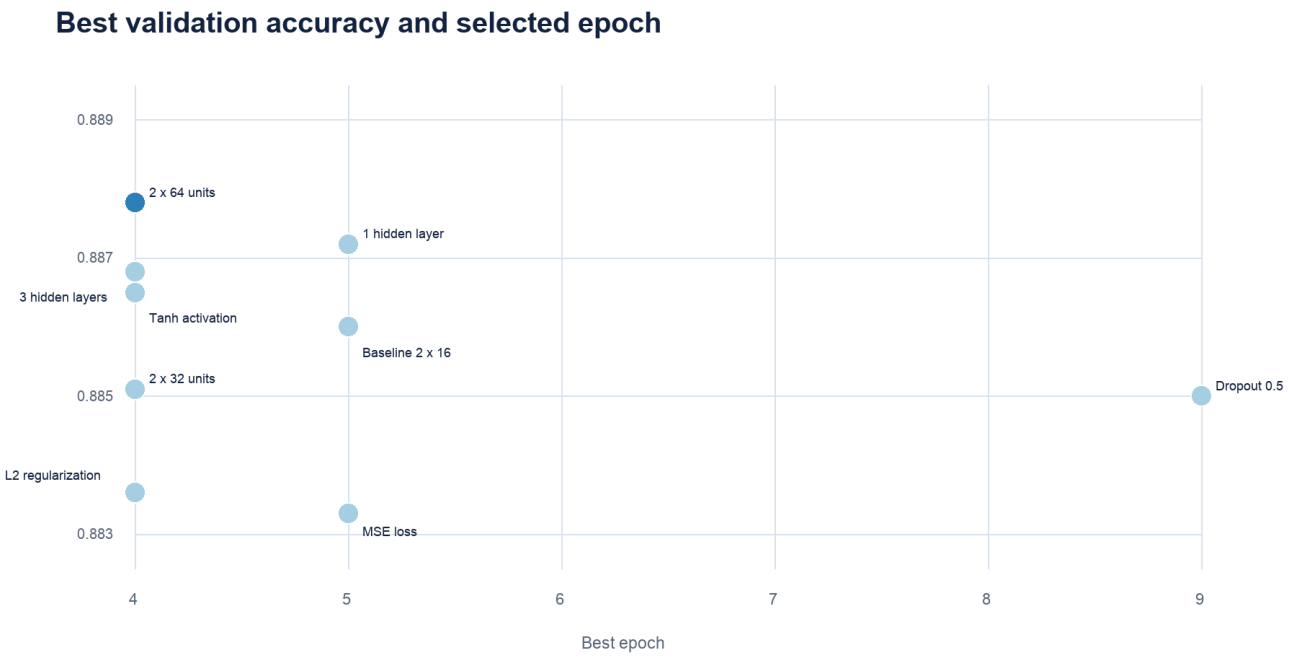


Controlled comparison of main design choices

Interpretation by experiment:

- **Hidden layers:** One hidden layer reached 0.8872, two layer baseline reached 0.8860, and three layers reached 0.8868. More layers did not create consistent improvement.
- **Hidden units:** Increasing both hidden layers to 64 units produced the best result. 32 unit network did not improve on 16-unit baseline.
- **Loss function:** Binary crossentropy outperformed MSE. This is also better conceptual match because model predicts probability for binary outcome.
- **Activation function:** Tanh produced a small improvement over ReLU, but difference was only 0.0005 and should not be overstated.
- **Regularisation:** Dropout and L2 regularization did not improve accuracy at tested strengths. These methods can still be useful, but their settings need additional tuning.

Training behavior and generalization:



Best validation accuracy and selected epoch

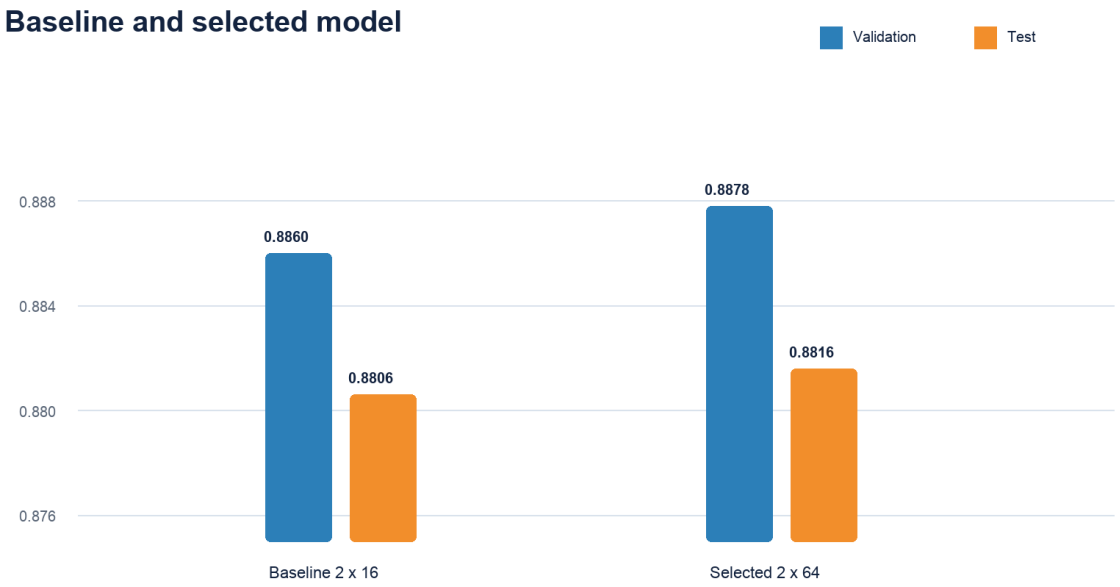
Most models reached their strongest validation accuracy around epoch 4 or 5. Dropout model trained until epoch 9 before early stopping selected its best result, but its validation accuracy remained below baseline. This is a reminder that training for more epochs does not necessarily enhance performance on new reviews.

Early stopping was used to retain weights from strongest validation epoch. This reduces the risk of overfitting, which occurs when a network learns details that are specific to its training data rather than patterns that transfer to unseen data. The training and validation curves in technical R Markdown report provide supporting evidence, while this summary focuses on decision and its meaning.

Validation and test relationship:

Validation results were used for model selection. Test results were reported only after each configuration had been defined. This distinction matters because repeatedly choosing a model based on test performance would gradually turn the test set into another tuning set and make final estimate less trustworthy.

Final model decision:



Baseline and selected model accuracy

Selected 2 x 64 model improved validation accuracy from 0.8860 to 0.8878. Test accuracy improved from 0.8806 to 0.8816. Test improvement was smaller than validation improvement, but it moved in same direction. That consistency supports selection while also showing that the practical gain is limited.

Recommended model specification:

Component	Selected setting
Input	10,000-position multi-hot review vector
Hidden architecture	Two dense layers with 64 units each
Hidden activation	ReLU
Output	One sigmoid unit
Optimizer	RMSprop
Loss	Binary crossentropy
Training control	Early stopping using validation accuracy
Selected epoch	Epoch 4

Decision:

I recommend 2 x 64 ReLU network. It provides highest observed validation accuracy and maintains small advantage on test set. I would not recommend adding more layers solely to increase complexity, and I would keep binary crossentropy rather than MSE.

Limitations and Conclusion:

Limitations

- **Small differences:** Best and baseline validation scores differ by only 0.0018. Another random split or training run could change the order of closely ranked models.
- **Simplified text representation:** Multi hot encoding records whether a word appears but ignores word order, repeated usage, and sentence structure.
- **Limited tuning range:** Only selected layer sizes, dropout, and L2 settings were tested. A different learning rate, dropout rate, or regularization strength could produce different results.
- **Accuracy alone:** Overall accuracy does not show which reviews were false positives or false negatives. Additional metrics would provide fuller evaluation.
- **Computing scope:** Work was designed as a controlled course experiment, not a production deployment study.

Conclusion:

Experiment met its objective by showing how several modeling choices affected performance on IMDB classification task. 2 x 64 ReLU network achieved the best validation accuracy and was selected as final model. Binary crossentropy remained strongest and most appropriate loss function. Additional layers and tested regularization settings did not automatically improve results.

Broader lesson is that model design should be based on controlled evidence rather than assumption that a larger or more complicated network will perform better. The final recommendation is supported by the validation and test results, but the small differences justify cautious language and further testing before any real world use.

Appendix results and terminology:

Model	Best epoch	Validation accuracy	Change from baseline
2 x 64 units	4	0.8878	+0.0018
1 hidden layer	5	0.8872	+0.0012
3 hidden layers	4	0.8868	+0.0008
Tanh activation	4	0.8865	+0.0005
Baseline 2 x 16	5	0.8860	+0.0000
2 x 32 units	4	0.8851	-0.0009
Dropout 0.5	9	0.8850	-0.0010
L2 regularization	4	0.8836	-0.0024
MSE loss	5	0.8833	-0.0027